



GLOBAL INSTITUTE OF TECHNOLOGY
B. Tech. III Semester Exam 2025-26
3CS-04/ DIGITAL ELECTRONICS (Guess Paper)
Branch CS/IT/AI&DS

Part A

- Q.1 Convert the following numbers (i) $(250.5)_{10} = ()_2$ (ii) $(101110.01)_2 = ()_8$
- Q.2 Which gates are called universal gates? What are its advantages?
- Q.3 Explain De Morgan's law.
- Q.4 Simplify the following expression $(A' + B)(A + B)$
- Q.5 Define Maxterm?
- Q.6 State and explain the basic law of Boolean logic operations.
- Q.7 Convert $(10110)_2$ to gray code.
- Q.8 Subtract using 2's complement method (-57) and (-33) .
- Q.9 Define Product of Sum with example.
- Q.10 Define Minterm.
- Q.11 Design full adder block diagram using two half adder.
- Q.12 Draw excitation table and state diagram of J-k Flip flop.
- Q.13 Design PMOS and NMOS inverter.
- Q.14 Why ECL logic is faster than TTL?
- Q.15 Compare between synchronous and Asynchronous counter.
- Q.16. What is master slave Flip-Flop
- Q.17 Define sequential and combinational circuit.
- Q.18. Define Race around condition
- Q.19 Explain the working of serial adder with suitable diagram.

- Q.20 Describe the serial in-parallel out shift register.
- Q.21 Why totem pole outputs cannot be connected together.
- Q.22 State advantage & disadvantage of TTL.
- Q.23 Explain Parallel adder with Block Diagram.
- Q.24 Show the excitation table of J-K Flip Flop.
- Q.25 Explain Half adder circuit. Also give its circuit Diagram.
- Q.26 How many 2:1 mux will be required to realize 128:1 mux? What is the use of select line in Mux.
- Q.27 Explain the short notes on Universal Shift Register.
- Q.28 What is edge triggered Flip-Flop? Explain the excitation table for D Flip-flop.
- Q.29 Draw the Symbol & truth Table of NAND & Ex-OR gate.
- Q.30 Differentiate Between Multiplexer & Demultiplexer.

Part B

- Q.1 Simplify the expression $[AB' + (ABC)' + A(B + AB')]'$
- Q.2 Convert the following function into canonical SOP form
 $f = AB + BC'D + A'D$
- Q.3 Subtract 835-274 using BCD subtraction.
- Q.4 Using K-Map simplify $f = (1, 3, 4, 5, 6, 7, 9, 12, 13)$
- Q.5 Write short notes on K-map & Tabular method.
- Q.6 Write short note on Encoder and Decoder.
- Q.7 Write short notes on (i) propagation delay time (ii) fan in (iii) fan out
 (iv) noise immunity (v) Power dissipation
- Q.8 Design 4 Bit binary Asynchronous UP-Down counter.
- Q.9 Explain NOT (inverter) gate with help of CMOS circuit diagram.

- Q.10 Design a 8 to 1 multiplexer by using the four variable function given by $F(A,B,C,D)=\sum(0,1,3,4,8,9,15)$
- Q.11 Draw a logic diagram of S-R flip-flop and obtain its characteristic equation. Also show its Excitation Table.
- Q.12 Explain the working of BCD adder.
- Q.13 Design the Mod-10 Counter.
- Q.14 Design BCD to Excess-3 code converter.
- Q.15 Design a 3 bit Synchronous Up/Down Counters.

Part C

- Q.1 Using the K-map method simplify the following the Boolean function and obtain (i) minimal SOP and (ii) minimal POS expression.
 $Y=\sum (0,2,3,6,7) +\sum d (8,10,11,15)$
- Q.2 Design all logic gate using universal gate.
- Q.3 Realize (a) $Y=A+BCD'$ using NAND gates. (b) $Y=(A+C) (A+D') (A+B+C')$ USING NOR Gates.
- Q.4
- Q.5 Draw the neat circuit of TTL NAND gate and explain.
- Q.6 Define Decoder. Explain BCD to seven segment decoder.
- Q.7 Simplify the logic function using Quine McCluskey minimization technique.
 $F(A,B,C,D)= \sum(1,3,5,8,9,11,15)+ \sum(2,13)$
- Q.8 Draw ECL (Emitter coupled logic)OR/NOR and explain it's working.
- Q.9 Discusses Cmos NAND and NOR gates.